

# Mohamed Elsayed

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## Education

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### Doctoral of Philosophy

University of Alberta, Canada  
Computing Science

**GPA: 4.0/4.0**

*Expected Graduation: Sep 2026*

### Master of Science

University of Alberta, Canada  
Computing Science [Thesis-based]

**GPA: 3.9/4.0**

*2019-2022*

### Bachelor of Science

University of science and technology, Zewail City, Egypt

**GPA: 3.7/4.0**

*2014 – 2019*

Major: Communication and Information Engineering. Minor: Physics of the Earth and Universe.

## Experience

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### Huawei Technologies Co., Ltd

Support Researcher, Intern

**Edmonton, Canada**

*May 2020 - Dec 2020*

- Worked on autonomous intersection navigation for the unprotected left turn problem with RL.
- Co-authored a workshop paper in NeuroIPS ML4AD.

### Nara Institute of Science and Technology

Robotics Research Intern

**Nara, Japan**

*Aug 2018 - Sep 2018*

- Worked on Semantic SLAM problem using deep learning techniques.
- Worked on map in-painting techniques.

### Valeo

Deep Learning Research Intern

**Smart Village, Egypt**

*Jul 2018 - Aug 2018*

- Worked on automotive-related projects (classification, semantic segmentation, and detection).
- Handled new state-of-the-art network architectures and bench-marked on related works.

### ONE Lab

Machine Learning Research Intern

**Cairo, Egypt**

*Jan 2018 - Mar 2018*

- Developed features that classify EEG signals into ictal and non-ictal signals for epilepsy patients.
- Published the results and findings in a conference paper bench-marking with related works.

## Teaching

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### University of Alberta

Graduate Teaching Assistant

**Edmonton, Canada**

*Sep 2019 - April 2020, Sep 2022 - Dec 2022*

- Taught two classes: introduction to programming and introduction to reinforcement learning.

### Zewail City of Science and Technology

Junior Teaching Assistant

**Cairo, Egypt**

*Sep 2018 - Dec 2018*

- Conducted presentations and assessed students in their assignments and problems in ML class.

## Selected Projects

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### RL Algorithms on Real-world Robots

Oct 2019 – Sep 2020

- Improved docking and moving behaviors.
- Produced robust docking in Roomba with online RL.

## Selected Publications

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### Conferences:

- **Elsayed, M.**, & Mahmood, A. R. (2024). Addressing loss of plasticity and catastrophic forgetting in continual learning. *International Conference on Learning Representations (ICLR)*.

### Workshops:

- **Elsayed, M.**, Mahmood, A. R. (2022) Utility-based Perturbed Gradient Descent: An Optimizer for Continual Learning. *NeurIPS Workshop on Optimization for Machine Learning*, 2022.
- **Elsayed, M.**, Mahmood, A. R. (2022) HesScale: Scalable Computation of Hessian Diagonals. *NeurIPS Workshop on Higher-Order Optimization in Machine Learning*, 2022.
- **Elsayed, M.**, Hassanzadeh, K., Nguyen, N. M., Alban, M., Zhu, X., Graves, D., & Luo, J. (2020). ULTRA: A reinforcement learning generalization benchmark for autonomous driving. *NeurIPS Workshop on Machine Learning for Autonomous Driving*.

## Extracurricular Activities

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- **Reviewer** at TMLR, ICLR 2023, NeurIPS 2022, CoLLAs 2024, RLC 2024.
- **President** of IEEE Zewail City Student Branch. (2017)  
Being in this position for a year has given me good experience in building and working with teams as a leader in addition to some managerial and planning skills (Granted University Best Leader Award).
- **Director** of Walking Robots Competition in Zewail City. (2017)